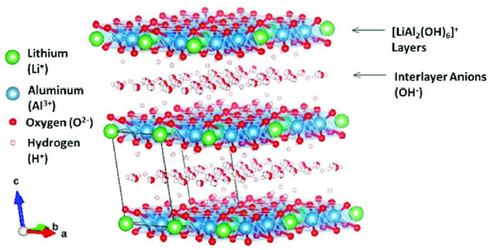
Direct lithium extraction from a geothermal brine & Theoretical capacity of hard carbon for sodium-ion batteries

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Abstract

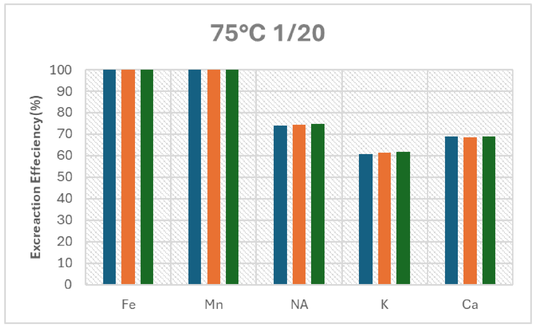
This project evaluates aluminum hydroxide [$Al(OH)_3$] as a host for lithium uptake from synthetic geothermal brine. The work identifies optimal reaction conditions, tests ion selectivity, and demonstrates lithium recovery.

1. Screen mass ratios and temperatures to find best efficiency (1/5, 1/10, 1/20, 1/40) (55°C and 75°C)
2. Test selectivity against competing ions under those conditions
3. Recover Li as LiCl from the layered double hydroxide product



Results

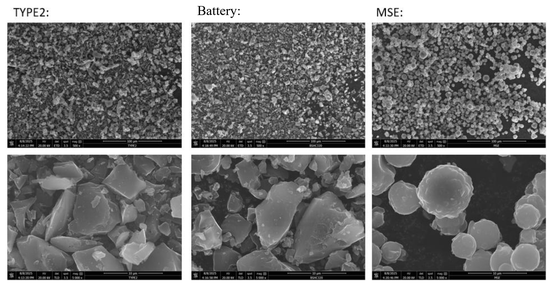
- No major differences were observed among the four mass ratios
- Ratios of 1/10 and 1/20 were selected for Stage 2 analysis
- Solid products are prepared for rinsing and ICP–OES analysis



Objective

This project evaluates the theoretical capacity of different particle shapes of commercially available hard carbons (HCs) for sodium-ion batteries (NIBs). The goal is to link particle shape to sodium storage behavior and guide future anode design.

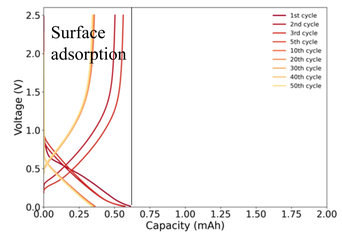
- Samples: TYPE and Battery (flake-like) vs. MSE (spherical)
- Aim: Relate structure to storage behavior and identify pathways for improved anodes



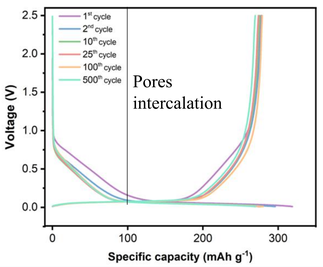
Electrochemical Results

- Flake carbons perform poorly at standard rates and degrade further with more cycles while spherical carbon retains strong capacity under practical charge/discharge conditions
- These results are based on 14 mm diameter disks. Even when binder type, concentration, or thickness are varied, flake carbons plateau near ~0.5 mAh, indicating that sodium storage is limited to surface interaction

Flake



Spherical



Procedure

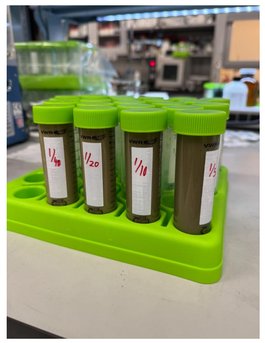
Step 1	Construct a Synthetic Brine
Step 2	Adjust Brine to pH 3 to pH 11
Step 3	Add Al(OH)3
Step 4	Heat until Dry
Step 5	Redisperse in Water and Centrifuge

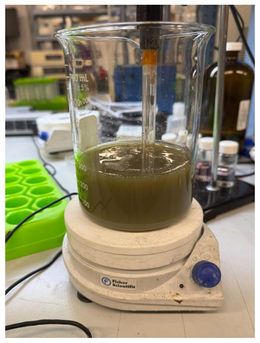
- Analyze by ICP–OES to compute Li extraction efficiency vs. temperature and ratio

Future

Next steps will confirm lithium selectivity and demonstrate recovery under practical conditions.

- Test selectivity using dual-salt brines
- Recover lithium from the LDH product by water rinsing and boehmite conversion
- Explore efficient scalable adaptations

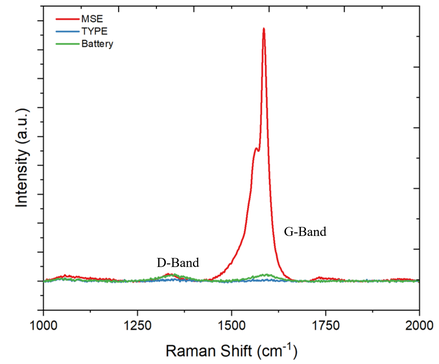




Physical Characterization

Flake carbons show poor carbon purity and low G/D band ratio, whereas spherical (MSE) carbons show excellent G band intensity.

- This demonstrates that spherical carbons have higher purity than flake carbons



Conclusion

- Particle shape is a key factor in sodium-ion storage, with spherical carbons showing the most favorable performance for future NIB anode development
- Spherical carbons support faster diffusion and exhibit higher stability while flake carbons remain constrained by sluggish ion transport and resistance

References

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[1] J. Teng, K. Zhang, B. Dai, L. Lei, S. Liu, T. Lu, J. Huang, F. Yang, H. Li, X. Tang, and J. Li, "Performance evaluation and failure analysis of $Na_{3.5}Fe_{2.5}(PO_4)_{1.5}P_2O_7$||hard carbon sodium-ion batteries: Implications for large-capacity full-cell configurations," *Energy*, vol. 323, p. 135844, 2025.

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